

Exercise class 2

1. Prove using the Fourier series of the sawtooth wave

$$f(x) = x, \quad -\pi < x < \pi, \quad f(x + 2\pi) = f(x)$$

and Parseval's identity that:

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

2. **Forced Oscillations under a Nonsinusoidal Periodic Driving Force**

Kreyszig textbook, Chapter 11.3, Example 1:

Show that if a_n and b_n are the Fourier coefficients of a function $f(x)$ then the Fourier coefficients of $f'(x)$ are given by $a'_0 = 0$, $a'_n = \frac{2\pi}{T} n b_n$ and $b'_n = -\frac{2\pi}{T} n a_n$.

What can we say about the coefficients of $\int f(x) dx$?

3. **Properties of the Fourier transform:**

a) Show that if $f(x)$ is a real function, then $\hat{f}_c(\omega) = \hat{f}_c(-\omega)$ and $\hat{f}_s(\omega) = -\hat{f}_s(-\omega)$

b) Show that $g(x) = f(-x)$ implies $\hat{g}_c(\omega) = \hat{f}_c(-\omega)$ and $\hat{g}_s(\omega) = \hat{f}_s(-\omega)$

c) Show that the Fourier transform of $g(x) = f'(x)$ is

$$\hat{g}_c(\omega) = \omega \hat{f}_s(\omega) \text{ and } \hat{g}_s(\omega) = -\omega \hat{f}_c(\omega)$$

d) Show that the Fourier transform is linear

e) Show that if of $g(x) = f(ax)$ ($a > 0$) then $\hat{g}_{c/s}(\omega) = \frac{1}{a} \hat{f}_{c/s}\left(\frac{\omega}{a}\right)$

4. **Calculation of Fourier transforms**

a) Cosine transform of

$$f(x) = \begin{cases} e^{-x} & \text{if } 0 < x \\ 0 & \text{if } x < 0 \end{cases}$$

b) Cosine transform of

$$g(x) = \begin{cases} e^{-ax} & \text{if } 0 < x \\ 0 & \text{if } x < 0 \end{cases}, \quad a > 0$$

c) $f(x) = e^{-|x|}$

d) Using that for $f(x) = e^{-x^2}$ the FT is $\hat{f}_c(\omega) = \sqrt{\pi} e^{-\frac{\omega^2}{4}}$ compute the FT of $g(x) = x e^{-x^2}$